Ryan Zhenqi Zhou, Ph.D. Candidate

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EDUCATION

University at Buffalo – SUNY, USA 09/2021 - 12/2025 (expected)

Ph.D. in Geographic Information System (STEM)

- **Geographic Information System Coursework:** GeoAI, Geospatial Big Data Analytics, Cartography, Remote Sensing
- **Data Science Coursework:** Machine Learning, Deep Learning, Time Series Modeling, Database, Data Visualization

SKILLS

- **Programming/Tools:** Python, R, SQL, Git, ArcGIS Pro
- **Libraries:** NumPy, Pandas, GeoPandas, SciPy, Scikit-Learn, Keras, TensorFlow, Matplotlib, Seaborn, Bokeh, Requests, JSON, PyQt
- **AI:** Regression (OLS, GWR, etc.), Tree-based Models (RF, XGBoost, GRF), Neural Networks (DNN, etc.), Clustering (K-means), Time Series Models (LSTM, Transformer, etc.), Explainable AI (SHAP), Generative AI (GPT)
- **Applications:** AI, Health, Climate Risk, Human Mobility, Big Data Analytics, GIScience, Remote Sensing

EXPERIENCE

Consultant, Global Health, World Bank - IFC, Washington, D.C. 05/2024 - Present

Healthcare, Climate, and Tourism Analysis and Modeling

- Processed, analyzed, and mapped large datasets, including healthcare facility data, patient records, climatic disaster data (floods, extreme heat, and droughts), tourism data, wealth index data, population data, and remote sensing images in low- and middle-income countries (LMICs) using **Python, ArcGIS Pro, and Map API**
- [AI] Implemented **linear regression (LR) and random forest (RF)** to predict wealth and population in Sub-Saharan Africa with accuracy over 95%. Leveraged **GPT API (Generative AI)** to match and standardize addresses
- [AI] Applied **XGBoost and Huff Model** to predict transactions of healthcare facilities and assess the existing market

Research Assistant, Weill Cornell Medical College, Cornell University, New York City 08/2024 - 06/2025

Real-World Evaluation of Cardiovascular Disease Risk Factors

- Collected and processed large datasets on cardiovascular disease, socioeconomic status, and demographic at the neighborhood level in New York City
- Matched patient addresses using fuzzy string matching and aggregated them to the neighborhood level for analysis and modeling
- Built **Python** workflows for scalable, reproducible data processing pipelines [GitHub repository]

Research Assistant, GeoAI Lab, University at Buffalo – SUNY, Buffalo 09/2021 - 05/2024

Climatic Disaster Impact Assessment

- Developed data-driven frameworks for assessing spatial and temporal impacts of climatic disasters (Texas Winter Storm and Buffalo Blizzard) using human mobility data, remote sensing images, and 311 service request data
- [AI] Applied **statistical models (SLM, SEM, and GWR), machine learning models (SVM, RF, and GRF), time series models (LSTM and Transformer), and explainable AI (SHAP)**, and then investigated impacts on different communities during climatic disasters [GitHub repository]
- First-author paper and paper published and presented at the 2023 and 2025 AAG Conferences

Obesity Estimation Optimization

- Processed human mobility data (3 GB), and derived diet and physical activity features at the neighborhood level using **Python**
- [AI] Developed **Geographical Random Forests (GRF) in Python** which takes spatial index into consideration [paper]
- [AI] Implemented **models (OLS, GWR, RF, DNN, and GRF)** to predict obesity prevalence with accuracy over 90% through cross validation [GitHub repository]
- First-author paper published and presented at the 2022 GEOMED Conference

Geo-Knowledge-Guided GPT Model

- [AI] Fused geo-knowledge with **GPT models (GPT-2 to GPT-4)** via prompt engineering and question-answering to extract and classify complex location descriptions with over 40% improvement in accuracy compared to NER baselines [paper]

Spatial Data Analyst (Intern), MetroDataTech, Shanghai 08/2020 - 09/2020

Educational Facility Assessment

- Evaluated the accessibility and land use of educational facilities in Shanghai using GeoPandas in **Python and ArcGIS Pro**
- Produced actionable insights that generated a $30K business profit

Research Assistant, Nanjing Forestry University, Nanjing 09/2018 - 06/2021

Medical Facility Accessibility Assessment

- Scraped travel distance and time records (2 billion) from communities to Wuhan medical facilities during the COVID-19 pandemic using **Python, SQL, and Map API**
- Assessed the accessibility of medical facilities and developed new service area plans to support decision-making
- First-author paper published and presented at the 2022 UCGIS Symposium